$k = 4$ in characteristic 2 past one word: a sparse matrix and $u128$ orders; IC reaches rho's cost near $m \approx 17$ (indicated, not established)

2026-10-15 to 2026-10-26. `research/gaudry_k4_char2_crossover_20261015/`. Protocol registered (commit `e403626cb`) before any claim run; binary sha256 `c143cf62…` for every claim session. Stacked on 1883 **Class: engineering** (Parts D and F) and a **comparison against rho on non-standard curves** (Part E).

What finished

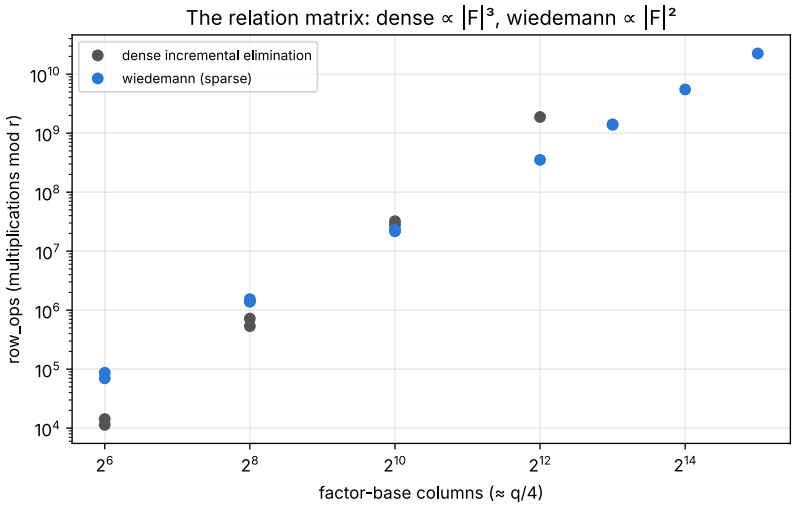
The container restarted repeatedly during the claim runs and each restart kills every run in flight (`ecbench` cannot resume a session). **Clean sessions:** Part D at $m = 8, 10, 12$ (18/18), Part E at $m = 15$ (6/6), the rho half of $m = 17$ (2/2), Part F (15/15). **Interrupted sessions:** $m = 16$ (3 of 4 runs), the IC half of $m = 17$ (1 of 2), Part D at $m = 14$ (2 of 3). Their audit passes `records`, `records_hash`, `plan_hash`, `spec_id`, `workloads`, `env_class` and `host_hash` and fails exactly `session_complete` and `record_count`; they were not replayed, so everything resting on them is *indicated*. **Not completed:** $m = 18$ (no attempt wrote a record). Part G (isolated hardware) was not run: no isolab credentials in this container. **None of this study's own sessions has been replayed** (the replay audits were stopped for lack of time, see the README's audit table); the integrity audits of the four complete sessions pass. The two replayed sessions are 1883's, re-executed with this binary (27/27 reproduced).

1. The matrix was the obstacle

In 1883 the dense matrix's `row_ops` grew as $|F|^3$ and would have reached rho's cost near $m = 18$. The `wiedemann` matrix (singleton-filtered core, Wiedemann on $R^T DR$, Berlekamp–Massey, exact check) grows as $|F|^2$:

m	columns	dense `row_ops`	sparse `row_ops`	dense ÷ sparse	trials, sparse ÷ dense
8	62	11,370; 14,252	69,871; 86,869	0.16; 0.16	1.0000
10	262	718,813; 536,268	1,536,771; 1,423,931	0.47; 0.38	1.0000
12	1,045	32.2 M; 28.0 M	22.2 M; 21.7 M	**1.45; 1.29**	1.0008; 1.0040
14	4,123	1.88 B (interrupted), 1.91 B (1883)	0.353 B (Part F)	**5.3–5.4**	1.0075; 1.0003

The sparse solver loses where the matrices are tiny and wins from $m = 12$. At $m = 17$ it is 1.4 % of the IC cost (interrupted session); it grows as $|F|$ relative to the relations, reaching 5 % near $m = 19$. The $m = 14$ row is unpaired (different seeds).

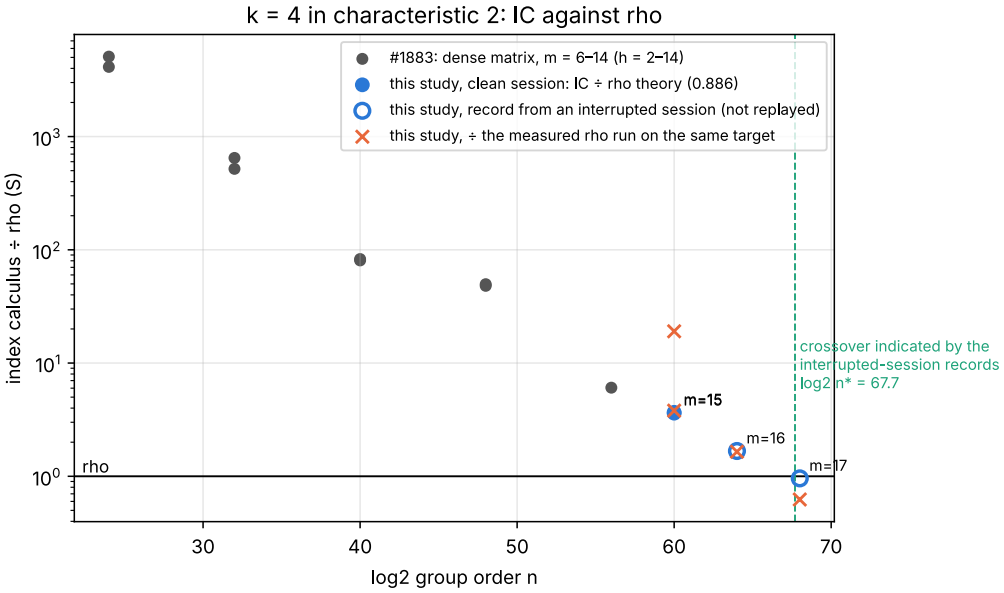


`row_ops` against factor-base columns. `raw/summary.json`.

2. IC against rho, cofactor $h = 2$, $m = 15$–17

m	$\log_2 n$	target	IC S	rho S	IC ÷ rho theory (0.886)	session
15	60	`W588ac37c7439`	3.198	0.168	**3.61**	clean
15	60	`W761fd6d9429f`	3.253	0.856	**3.67**	clean
16	64	`Wa1b3b10e6968`	1.482	0.899	*1.67*	interrupted
17	68	`Wd5f7dba84d0e`	0.849	1.361	*0.96*	IC interrupted, rho clean

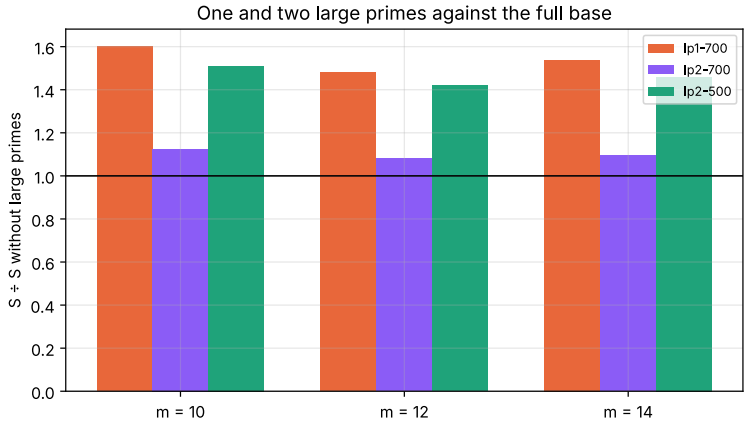
At every size IC used 191–196 trials per relation. **Established:** IC is 3.6× rho's theoretical cost at $m = 15$. **Indicated** (one target each, interrupted sessions): 1.67× at $m = 16$ and 0.96× at $m = 17$; the fit through the three points has slope -0.24 ($-\frac{1}{4}$ expected) and crosses 1 at $\log_2 n^* \approx 67.7$ ($m^* \approx 16.9$), against 1883's extrapolated $2^{68.1}$. **Not shown:** that IC beats rho. One target at $m = 17$ from an incomplete session, at 0.96× rho's *theoretical* cost and inside rho's run-to-run spread, is not that; the registered claim (E3) was at $m = 18$, which did not run.



IC ÷ rho against $\log_2 n$: 1883 (grey, dense matrix, $h = 2$–14), clean-session points (filled), interrupted-session points (hollow), IC ÷ the measured rho run (crosses). `raw/summary.json`.

3. Large primes lose at these sizes (Part F, clean)

S ÷ the same run without large primes: one large prime over 70 % of the columns 1.60, 1.48, 1.54; two large primes over 70 %: 1.13, 1.08, 1.10; over 50 %: 1.51, 1.42, 1.46 ($m = 10, 12, 14$). The matrix they shrink is under 1 % of the cost (its cost falls to 0.52–0.78×); merging and the extra trials cost more. By the protocol's model the variant pays near $n \approx 2^{92}$. A recorded negative at every reachable size.



$S \div S$ without large primes. `raw/summary.json`.

Predictions and checks

D1 *not met as registered* (a spec error: the measurement seed differs from 1883's, so the dense arms are fresh samples; the pilot with the seed reused matched exactly, and this binary replays all 27 committed 1883 runs, `audits/reaudit-*`). D2, D3, F1, F2: held. E2: held at $m = 15$, indicated at 16 and 17. E5: indicated (16.9). E3, E4: not tested ($m = 18$). Frobenius parity: even on 375/375 curves ($m = 4$–18, 25 each), measured, not proved.

Scope. Curves over $\mathbb{F}_{2^{4m}}$ are not standard curves. The established result is a trend through $m = 15$; the crossover and the $m = 18$ comparison are open. Re-running `run_claims.sh` on a host that does not restart completes them (about 15 hours for $m = 17$ IC).

Environment: Claude Code cloud container, Xeon @ 2.10 GHz, 4 vCPU, four sessions at a time on separate cores (L1). Conductor task T-4.